

SAT Math Mastery · Week 5A

Exponentials & Sequences — Answer Key & Worked Solutions

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics exponential growth and decay models · base and initial value · doubling and half-life · compound growth · exponent laws · arithmetic vs geometric sequences · linear vs exponential · tables and graphs

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Build $y = ab^t$ or $y = a(1 \pm r)^{t/k}$ from a verbal description (percent $\rightarrow$ factor; “every k units” $\rightarrow$ exponent t/k).
- S2.** Read the initial value and the growth or decay factor from an equation, a table or a graph.
- S3.** Use doubling time and half-life; solve for a time by matching powers of the same base.
- S4.** Tell exponential from linear: constant *ratio* versus constant *difference*.
- S5.** Apply the exponent laws, rational exponents and equal-base equations; rewrite equivalent exponential forms.
- S6.** Arithmetic sequences $a_n = a_1 + (n-1)d$ and geometric sequences $a_n = a_1r^{n-1}$ (including negative ratios).
- S7.** Interpret constants and factors in context; identify the growth type from “equals $p\%$ of the previous value”.
- S8.** Build a and b from two points; use parameters, shifts and hidden conditions.

2 · REQUIRED FACTS

- Exponential: equal steps in x multiply y by the same factor. Linear: equal steps in x add the same amount to y .
- $y = ab^x$ with $a > 0$: $b > 1$ is growth, $0 < b < 1$ is decay; a is the value at $x = 0$ and the graph never touches the x -axis.
- A change of $p\%$ per period has factor $1 + \frac{p}{100}$ (growth) or $1 - \frac{p}{100}$ (decay), never $\frac{p}{100}$ alone.
- “Every k units” puts $\frac{t}{k}$ in the exponent; the factor for one unit is the k -th root of the factor for k units.
- The exponent in a geometric sequence is $n - 1$, not n . Terms m apart differ by the factor r^m (or by md in an arithmetic sequence).
- A compounded balance over n periods is $P(1 + \frac{r}{n})^{nt}$ for annual rate r , n periods per year.

3 · REQUIRED FORMULAS

$$y = ab^x, \quad y = a(1 \pm r)^t, \quad y = ab^{t/k}, \quad a_n = a_1r^{n-1}, \quad a_n = a_1 + (n-1)d$$

$$b^m b^n = b^{m+n}, \quad \frac{b^m}{b^n} = b^{m-n}, \quad (b^m)^n = b^{mn}, \quad b^{-n} = \frac{1}{b^n}, \quad b^{m/n} = \sqrt[n]{b^m}, \quad b^0 = 1$$

$$\text{two points: } b^{x_2-x_1} = \frac{y_2}{y_1}, \quad A = P \left(1 + \frac{r}{n}\right)^{nt}, \quad \text{half-life: } N = N_0 \left(\frac{1}{2}\right)^{t/h}$$

4 · MINI THEORY SHEET

Linear or exponential?

	Linear	Exponential
Table	equal Δy for equal Δx	equal y_2/y_1 for equal Δx
Words	“increases by c each period”	“increases by $p\%$ each period”
Equation	$y = mx + b$	$y = ab^x$
Long run	grows by the same amount	eventually beats every linear model

Reading $y = ab^t$. a = value when $t = 0$. b = factor per unit of t ; the percent change is $(b - 1) \cdot 100\%$. If the exponent is t/k , then b is the factor per k units.

Rewriting the base. $b^{t/k} = (b^{1/k})^t$. Write both sides over one common base (powers of 2, 3, 5) and compare exponents.

From a table or two points. Divide the outputs and take a root of the step count: if x moves by m , then $b^m = \frac{y_2}{y_1}$. Then $a = y_1 b^{-x_1}$.

Sequences. Same ratio $\Rightarrow$ geometric, same difference $\Rightarrow$ arithmetic. Three consecutive terms p, q, r of a geometric sequence satisfy $q^2 = pr$; of an arithmetic sequence, $2q = p + r$.

Context. The constant out front is the starting amount; the base is the multiplier per period; never read the exponent as the percent.

Speed shortcuts.

Situation	Shortcut
doubling / halving	count how many doublings: 2^n ; target = start $\times 2^n$
percent $\rightarrow$ factor	$+7\% \rightarrow 1.07$, $-12\% \rightarrow 0.88$; multiply by the factor n times
every k units	exponent $\frac{t}{k}$; rewrite as $(b^{1/k})^t$ for a per-unit factor
solve for an expression	factor out 3^x (e.g. $3^{x+2} - 3^x = 8 \cdot 3^x$) and read off the block
two table rows	$b^{\Delta x} = \frac{y_2}{y_1}$, then work back to a with one equation
geometric three-in-a-row	$q^2 = pr$ gives the unknown immediately
linear vs exponential race	test values; the exponential only wins after a crossover month
shift in a graph	$f(x - h) = a b^{-h} b^x$: the shift changes only the leading constant

5 · GUIDED EXAMPLES

Example 1 — Build the model from words

A medicine concentration starts at 240 mg/L and falls by 30% every 4 hours. Write $C(t)$, t in hours, and find $C(8)$.

Factor per 4 hours: $1 - 0.30 = 0.70$. So $C(t) = 240(0.7)^{t/4}$.

$C(8) = 240(0.7)^2 = 240(0.49) = 117.6$ mg/L.

Example 2 — Two points $\rightarrow a$ and b

An exponential function has $f(1) = 12$ and $f(4) = 96$. Find $f(x)$ and $f(0)$.

Three steps from $x = 1$ to $x = 4$: $b^3 = \frac{96}{12} = 8 \Rightarrow b = 2$.

$12 = a(2)^1 \Rightarrow a = 6$. So $f(x) = 6(2)^x$ and $f(0) = 6$.

Example 3 — Match the base

Solve $4^{x+1} = 32^{x-2}$.

$4 = 2^2$ and $32 = 2^5$: $2^{2(x+1)} = 2^{5(x-2)}$. Exponents: $2x + 2 = 5x - 10 \Rightarrow x = 4$.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3 \cdot 2^x$. What is the value of $f(5) - f(3)$?

Worked solution

1. $f(5) = 3 \cdot 2^5 = 3 \cdot 32 = 96.$

2. $f(3) = 3 \cdot 2^3 = 3 \cdot 8 = 24.$

3. $f(5) - f(3) = 96 - 24 = 72.$

Answer: 72

2.

EASY · NO CALCULATOR NEEDED

The table shows values of the function f for selected values of x . Which of the following defines f ?

x	0	1	2	3
$f(x)$	64	48	36	27

(A) $f(x) = 64 \left(\frac{1}{4}\right)^x$

(B) $f(x) = 64 \left(\frac{3}{4}\right)^x \checkmark$

(C) $f(x) = 64 - 16x$

(D) $f(x) = 64 \left(\frac{4}{3}\right)^x$

Worked solution

1. Each output is the previous one times the same factor: $\frac{48}{64} = \frac{36}{48} = \frac{27}{36} = \frac{3}{4}.$

2. The differences $(-16, -12, -9)$ are not constant, so the function is not linear.

3. With $f(0) = 64$ and ratio $\frac{3}{4}$: $f(x) = 64 \left(\frac{3}{4}\right)^x.$

Answer: (B)

3.

EASY · NO CALCULATOR NEEDED

The balance, in dollars, of a savings account t years after the account is opened is modeled by $A(t) = 800(1.05)^t$. Which of the following statements best describes the account?

(A) Starts at \$800; up 105% per year.

(B) Starts at \$1.05; up \$800 per year.

(C) Starts at \$800; up 5% per year. $\checkmark$

(D) Starts at \$800; up \$5 per year.

Worked solution

1. In $A(t) = ab^t$, $a = 800$ is the starting balance and $b = 1.05$ is the yearly factor.2. A factor of $1.05 = 1 + 0.05$ means the balance grows by 5% each year (not by \$5, and not by 105%).**Answer:** (C)

4.

EASY · NO CALCULATOR NEEDED

The first term of a geometric sequence is 6 and the common ratio is $\frac{1}{2}$. What is the fourth term of the sequence?

(A) $\frac{3}{4}$ ✓

(B) $\frac{3}{8}$

(C) $\frac{3}{2}$

(D) $\frac{3}{16}$

Worked solution

1. $a_n = a_1 r^{n-1}$, so $a_4 = 6 \left(\frac{1}{2}\right)^3$.

2. $a_4 = 6 \cdot \frac{1}{8} = \frac{3}{4}$.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{(2x^3)^3}{4x^2}$, for $x \neq 0$?

(A) $\frac{x^7}{2}$

(B) $2x^{11}$

(C) $2x^{\frac{9}{2}}$

(D) $2x^7$ ✓

Worked solution

1. Cube everything inside: $(2x^3)^3 = 2^3 x^9 = 8x^9$.

2. Divide: $\frac{8x^9}{4x^2} = 2x^{9-2} = 2x^7$.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

A lab culture contains 500 bacteria at time $t = 0$, where t is measured in hours, and the number of bacteria doubles every 3 hours. Which of the following equations gives the number of bacteria N at time t ?

(A) $N = 500(2)^t$

(B) $N = 500(2)^{3t}$

(C) $N = 500(2)^{t/3}$ ✓

(D) $N = 500(3)^t$

Worked solution

1. Doubling every 3 hours means t hours contain $\frac{t}{3}$ doublings.

2. $N = 500 \cdot 2^{t/3}$. Check at $t = 3$: $500 \cdot 2^1 = 1000$ (doubled once).

3. The other options fail the same check: $500(2)^3 = 4000$, $500(2)^9 = 256,000$ and $500(3)^3 = 13,500$ at $t = 3$.

Answer: (C)

7.

MEDIUM · NO CALCULATOR NEEDED

A 640-milligram sample of a radioactive substance loses half of its mass every 5 days. After how many days will exactly 20 milligrams of the sample remain?

Worked solution

1. Halve repeatedly: $640 \rightarrow 320 \rightarrow 160 \rightarrow 80 \rightarrow 40 \rightarrow 20$.

2. That is 5 halvings, and each takes 5 days.

3. $5 \times 5 = 25$ days. (Equivalently $640 \left(\frac{1}{2}\right)^{t/5} = 20 \Rightarrow 2^{t/5} = 32 = 2^5 \Rightarrow t = 25$.)

Answer: 25

8.

MEDIUM · CALCULATOR

A delivery van is bought for \$28,000 and loses 12% of its value each year. To the nearest dollar, what is the value of the van 3 years after it was purchased?

(A) \$17,920

(B) \$19,081 ✓

(C) \$24,640

(D) \$39,338

Worked solution

1. Each year the value is multiplied by $1 - 0.12 = 0.88$.
2. $V(3) = 28,000(0.88)^3 = 28,000(0.681472) = 19,081.2$.
3. To the nearest dollar: \$19,081. (Subtracting 12% of the original price three times gives \$17,920, which treats the decay as linear.)

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

The exponential function f satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(0)$?

Worked solution

1. From $x = 2$ to $x = 5$ is 3 steps: $b^3 = \frac{486}{18} = 27 \Rightarrow b = 3$.
2. $f(x) = a(3)^x$ and $f(2) = 18$: $9a = 18 \Rightarrow a = 2$.
3. $f(0) = a = 2$.

Answer: 2

10.

MEDIUM · NO CALCULATOR NEEDED

The first term of a geometric sequence is 5 and the fourth term is 135. What is the second term?

(A) 27

(B) 45

(C) 70

(D) 15 ✓**Worked solution**

1. $a_4 = a_1 r^3 \Rightarrow 135 = 5r^3 \Rightarrow r^3 = 27 \Rightarrow r = 3$.
2. $a_2 = a_1 r = 5 \cdot 3 = 15$.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

Each year, the population of a town is 94% of its population the previous year. Which of the following types of function best models the town's population as a function of the number of years?

(A) Exponential decay ✓

(B) Increasing linear

(C) Decreasing linear

(D) Exponential growth

Worked solution

1. "Is 94% of the previous value" is a multiplication by the same factor 0.94 each year, so the model is exponential, not linear.
2. The factor $0.94 < 1$, so the population shrinks: exponential decay.

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The value, in dollars, of a collectible t years after 2020 is modeled by $V(t) = 1800(0.92)^t$. Which of the following is the best interpretation of the number 0.92 in this context?

(A) The value in 2020 was \$0.92.

(B) The value decreases by 8% each year. ✓

(C) The value decreases by 92% each year.

(D) The value decreases by \$0.92 each year.

Worked solution

- 0.92 is the factor applied each year: each year's value is 92% of the previous year's.
- A factor of $0.92 = 1 - 0.08$ is a decrease of 8% per year (not 92%, and not a \$0.92 drop).

Answer: (B)

13.

MEDIUM · NO CALCULATOR NEEDED

If $27^{x-1} = 9^{x+2}$, what is the value of x ?

Worked solution

- Write both sides with base 3: $27 = 3^3$ and $9 = 3^2$.
- $3^{3(x-1)} = 3^{2(x+2)} \Rightarrow 3x - 3 = 2x + 4$.
- $x = 7$. (Check: $27^6 = 3^{18} = 9^9$.)

Answer: 7

14.

HARD · CALCULATOR

A savings account is modeled by $A(t) = 5000(1.015)^{4t}$, where $A(t)$ is the balance in dollars t years after the initial deposit. Which of the following is closest to the percent by which the balance increases each year?

(A) 1.50%

(B) 6.00%

(C) 6.14% ✓

(D) 15.0%

Worked solution

- The exponent $4t$ means the factor 1.015 is applied 4 times per year.
- Yearly factor: $(1.015)^4 \approx 1.0614$.
- The yearly increase is about 6.14% (not 6%, which ignores compounding, and not 1.5%, which is per quarter).

Answer: (C)

15.

HARD · NO CALCULATOR NEEDED

The function f is defined by $f(t) = 90 \cdot 8^{t/6}$. Which of the following functions is equivalent to f ?

(A) $90(\sqrt{2})^t$ ✓(B) $90 \cdot 2^{t/6}$ (C) $90\left(\frac{4}{3}\right)^t$ (D) $90 \cdot 4^{t/6}$ **Worked solution**

- Rewrite $8 = 2^3$: $8^{t/6} = 2^{3t/6} = 2^{t/2}$.
- $2^{t/2} = (2^{1/2})^t = (\sqrt{2})^t$, so $f(t) = 90(\sqrt{2})^t$.
- Check at $t = 6$: $90 \cdot 8 = 720$ and $90(\sqrt{2})^6 = 90 \cdot 8 = 720$.

Answer: (A)

16.

HARD · CALCULATOR

For the exponential function $f(x) = a \cdot b^x$, where a and b are positive constants, $f(1) = 24$ and $f(4) = 81$. What is the value of $f(6)$?

(A) 243

(B) $\frac{2187}{8}$ (C) $\frac{243}{2}$ (D) $\frac{729}{4}$ ✓**Worked solution**

1. Three steps from $x = 1$ to $x = 4$: $b^3 = \frac{81}{24} = \frac{27}{8} \Rightarrow b = \frac{3}{2}$.

2. $24 = a \cdot \frac{3}{2} \Rightarrow a = 16$, so $f(x) = 16 \left(\frac{3}{2}\right)^x$.

3. $f(6) = 16 \cdot \frac{729}{64} = \frac{729}{4} = 182.25$. (Shortcut: $f(6) = f(4) \cdot b^2 = 81 \cdot \frac{9}{4} = \frac{729}{4}$.)

Answer: (D)

17.

HARD · DESMOS

Channel A has 50 subscribers and gains 30 subscribers each month. Channel B has 50 subscribers and its subscriber count increases by 20% each month. What is the least whole number of months after which Channel B has more subscribers than Channel A?

Worked solution

1. A: $50 + 30m$. B: $50(1.2)^m$. Both start at 50.

2. For small m the linear channel leads (e.g. $m = 6$: 230 vs ≈ 149 ; $m = 10$: 350 vs ≈ 309.6 ; $m = 11$: 380 vs ≈ 371.5).

3. $m = 12$: A has 410 and B has $50(1.2)^{12} \approx 445.9$. B first exceeds A at $m = 12$.

Answer: 12

18.

HARD · NO CALCULATOR NEEDED

The second term of a geometric sequence is 6, the fourth term is 54, and the common ratio is negative. What is the fifth term?

Worked solution

1. $a_4 = a_2 r^2 \Rightarrow 54 = 6r^2 \Rightarrow r^2 = 9$.

2. The ratio is negative, so $r = -3$.

3. $a_5 = a_4 \cdot r = 54(-3) = -162$.

Answer: -162

19.

HARD · CALCULATOR

A radioactive sample loses 36% of its mass every 5 days. Its mass can also be modeled by $M(t) = M_0 b^t$, where t is the number of days and b is a constant. What is the value of b , rounded to the nearest thousandth?

Worked solution

1. Every 5 days the mass is multiplied by $1 - 0.36 = 0.64$: $b^5 = 0.64$.

2. $b = 0.64^{1/5} \approx 0.9146$.

3. Rounded to the nearest thousandth: 0.915.

Answer: 0.915

20.

HARD · NO CALCULATOR NEEDED

If $3^{x+2} - 3^x = 216$, what is the value of 3^{x+1} ?

Worked solution

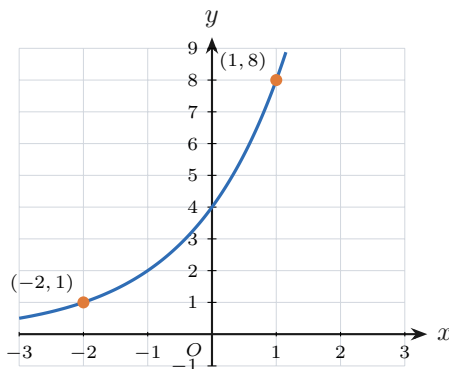
- Factor out 3^x : $3^{x+2} - 3^x = 3^x(9 - 1) = 8 \cdot 3^x$.
- $8 \cdot 3^x = 216 \Rightarrow 3^x = 27$.
- $3^{x+1} = 3 \cdot 3^x = 3 \cdot 27 = 81$.

Answer: 81

21.

HARD · DESMOS

The graph of the function $f(x) = a \cdot b^x$, where a and b are positive constants, is shown in the xy -plane. The points $(-2, 1)$ and $(1, 8)$ lie on the graph. What is the value of $a + b$?



(A) 10

(B) 12

(C) 6 ✓

(D) 8

Worked solution

- Read the two marked points: $f(-2) = 1$ and $f(1) = 8$.
- Three steps from $x = -2$ to $x = 1$: $b^3 = \frac{8}{1} = 8 \Rightarrow b = 2$.
- $f(1) = a \cdot 2 = 8 \Rightarrow a = 4$ (check: $4 \cdot 2^{-2} = 1$).
- $a + b = 4 + 2 = 6$.

Answer: (C)

22.

HARD · NO CALCULATOR NEEDED

The numbers $x + 2$, $x + 8$ and $x + 26$ are, in that order, three consecutive terms of a geometric sequence. What is the value of x ?

(A) 6

(B) 1 ✓

(C) 2

(D) 4

Worked solution

- In a geometric sequence the middle term squared equals the product of its neighbors: $(x+8)^2 = (x+2)(x+26)$.
- $x^2 + 16x + 64 = x^2 + 28x + 52 \Rightarrow 12 = 12x \Rightarrow x = 1$.
- Check: 3, 9, 27 with ratio 3.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

For positive x and y , the expression $\frac{(x^{2/3}y^{-1})^3}{x^{-1}y^2}$ is equivalent to x^py^q , where p and q are constants. What is the value of $p + q$?

(A) 0

(B) 8

(C) -8

(D) -2 ✓

Worked solution1. Numerator: $(x^{2/3}y^{-1})^3 = x^2y^{-3}$.2. Divide by $x^{-1}y^2$ (subtract exponents): $x^{2-(-1)}y^{-3-2} = x^3y^{-5}$.3. $p = 3$, $q = -5$, so $p + q = -2$.**Answer: (D)**

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

An arithmetic sequence a_n and a geometric sequence g_n both have first term 2. The common difference of a_n and the common ratio of g_n are both equal to k , where $k > 1$. If $g_3 - a_3 = 38$, what is the value of g_5 ?

Worked solution1. $a_3 = 2 + 2k$ and $g_3 = 2k^2$.2. $2k^2 - (2 + 2k) = 38 \Rightarrow 2k^2 - 2k - 40 = 0 \Rightarrow k^2 - k - 20 = 0 \Rightarrow (k - 5)(k + 4) = 0$.3. Since $k > 1$, $k = 5$.4. $g_5 = 2 \cdot 5^4 = 2 \cdot 625 = 1250$.**Answer: 1250**

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For the function $f(x) = a \cdot b^x$, where a and b are constants and $b > 0$, $f(2) = 12$ and $f(4) - f(2) = 96$. What is the value of $f(5)$?

Worked solution1. $f(4) = f(2)b^2 = 12b^2$, so $f(4) - f(2) = 12b^2 - 12 = 96$.2. $b^2 = 9$; since $b > 0$, $b = 3$.3. $f(5) = f(2)b^3 = 12 \cdot 27 = 324$. (Also $a = \frac{12}{9} = \frac{4}{3}$ and $f(5) = \frac{4}{3} \cdot 243 = 324$.)**Answer: 324**

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

Culture A starts with 4000 cells and its population is cut in half every 2 hours. Culture B starts with 500 cells and its population doubles every 3 hours. After how many hours, t , will the two cultures have the same number of cells?

(A) 3.6 ✓

(B) 18

(C) 0.6

(D) 1.8

Worked solution1. A: $4000 \left(\frac{1}{2}\right)^{t/2} = 4000 \cdot 2^{-t/2}$. B: $500 \cdot 2^{t/3}$.2. Set equal and divide by 500: $8 \cdot 2^{-t/2} = 2^{t/3} \Rightarrow 2^{3-t/2} = 2^{t/3}$.3. $3 - \frac{t}{2} = \frac{t}{3} \Rightarrow 3 = \frac{5t}{6} \Rightarrow t = \frac{18}{5} = 3.6$ hours.**Answer: (A)**

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

The terms of a geometric sequence are $a_1, a_2, a_3, \dots$. The sum of the second and third terms is 12, and the sum of the third and fourth terms is 36. What is the value of a_6 ?

Worked solution

1. $a_3 + a_4 = r(a_2 + a_3)$, so $r = \frac{36}{12} = 3$.
2. $a_2 + a_3 = a_2(1 + 3) = 12 \Rightarrow a_2 = 3$, so $a_1 = 1$.
3. $a_6 = a_1 r^5 = 3^5 = 243$.

Answer: 243

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A quantity decreases by 19% every 2 years. Its current value is 4000. In how many years from now will its value be 2916?

Worked solution

1. Every 2 years the factor is $1 - 0.19 = 0.81 = 0.9^2$, so the yearly factor is 0.9.
2. $V(t) = 4000(0.9)^t$. Set $4000(0.9)^t = 2916 \Rightarrow 0.9^t = 0.729$.
3. $0.729 = 0.9^3$, so $t = 3$ years. (Check: $4000 \cdot 0.729 = 2916$.)

Answer: 3

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. If $f(3) > f(2)$, which of the following statements must be true?

I. $f(-1) < f(0)$ II. $f(5) - f(4) > f(4) - f(3)$ III. $f(0) > 1$

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II and III

Worked solution

1. $f(3) > f(2) \iff ab^3 > ab^2 \iff b > 1$ (since $a, b > 0$). The function is increasing.
2. I: increasing, so $f(-1) < f(0)$. True.
3. II: $f(5) - f(4) = ab^4(b - 1)$ and $f(4) - f(3) = ab^3(b - 1)$; the first is b times the second and $b > 1$, so it is larger. True.
4. III: only $a > 0$ is known; for example $a = \frac{1}{2}$, $b = 2$ gives $f(0) = \frac{1}{2} < 1$. Not necessarily true.
5. Only I and II must be true.

Answer: (B)

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 48 \left(\frac{2}{3}\right)^x$. The function g is defined by $g(x) = f(2x - 4)$ and can be written in the form $g(x) = c \cdot d^x$, where c and d are constants. What is the value of $c \cdot d$?

(A) 243

(B) 48

(C) 108 ✓

(D) 162

Worked solution

1. $g(x) = 48 \left(\frac{2}{3}\right)^{2x-4} = 48 \left(\frac{2}{3}\right)^{-4} \left[\left(\frac{2}{3}\right)^2\right]^x$.

2. $d = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$ and $c = 48 \left(\frac{2}{3}\right)^{-4} = 48 \cdot \frac{81}{16} = 243$.

3. $c \cdot d = 243 \cdot \frac{4}{9} = 108$.

Answer: (C)